

Year 6 Science Bridging Programme



Project 1

THE GREAT SCIENCE INVESTIGATOR

Inspire Academic™
Learning how science works

Vol1.0

INSPIRE ACADEMIC™

YEAR 6 SCIENCE BRIDGING PROGRAMME

PROJECT 1 WORKBOOK

THE GREAT SCIENCE INVESTIGATOR

Welcome, Young Scientist

Science is about **finding reliable answers**.
Science is not about guessing.

In this project, you will learn how scientists investigate questions about the world. You will plan your own investigation, collect real data, and explain what your results show.

By the end of this project, you won't just *do* experiments,
you will **think like a scientist**.

PROJECT BIG QUESTION

How do scientists find reliable answers to questions about the world?

Keep this question in mind as you work through this project.

WHAT YOU WILL LEARN TO DO

By the end of Project 1, you will be able to:

- ask scientific questions
- plan a fair test
- identify variables
- measure and record results
- draw graphs
- explain conclusions using evidence and suggest improvements to an investigation

Preface

Science is one of the most powerful ways human beings have learned to understand the world.

It teaches us to look closely, to ask careful questions, to test ideas fairly, and to trust evidence rather than guesses. At its best, science trains the mind to be curious, patient, and honest.

This workbook is not about memorising facts or rushing to the “right answer”.

It is about learning **how science works**.

In *Project 1: The Great Science Investigator*, you will plan and carry out your own investigation, collect real data, draw graphs, and explain what your results show.

Along the way, you will learn the habits that scientists use every day — habits that will help you not only in science, but in thinking clearly in all areas of learning.

Take your time. Work carefully. Be curious.

By the end of this project, you will not just have completed an experiment — you will have taken your first steps into **thinking like a scientist**.

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London – 2026

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SECTION 1

WHAT MAKES A QUESTION SCIENTIFIC?

Not all questions can be answered by science.

A **scientific question**:

- can be tested
- can be measured
- can be answered using evidence

Activity 1.1 — Question Sorting

Read the questions below. Tick ✓ the ones that are scientific.

Question	Scientific?
Does a plant grow taller in sunlight?	☐
What is the best football team?	☐
Does sugar dissolve faster in hot water?	☐
Why do people like music?	☐

Activity 1.2 Make It Scientific

Rewrite each question so it becomes scientific.

1. Which shoes are better?

2. Which paper plane is best?

Reflection

Why do scientists need questions that can be tested?



SECTION 2

VARIABLES & FAIR TESTING

In science, we change **one thing at a time**.

Key Words

- **Independent variable** — what you change
- **Dependent variable** — what you measure
- **Control variables** — what you keep the same
- **Fair test** — only one thing is changed

Activity 2.1 — Spot the Variables

Question:

Does the height a ball is dropped from, affect how high it bounces?

Fill in the table:

Type of Variable	What is it?
Independent variable	_____
Dependent variable	_____
One control variable	_____

Activity 2.2 — Is This Fair?

A pupil tests how far a toy car rolls but:

- uses different cars
- changes the ramp height each time

Is this a fair test? Why or why not?



SECTION 3

PLANNING YOUR INVESTIGATION

Now you will plan your own scientific investigation.

Investigation Title



Aim (What are you trying to find out?)



Variables

Type	Write yours here
Independent variable	
Dependent variable	
Control variables	

Prediction (*Your hypothesis*)

I predict that:



Method (What will you do?)

Number your steps clearly.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____

Equipment Needed

(list, draw, sketch etc...)



SECTION 4 — RESULTS & DATA

Results Table

e.g

Trial	Measurement 1 (cm)	Measurement 2 (cm)	Average Measurement (cm)
1			
2			
3			

Repeat & Average

Why do scientists repeat measurements?



SECTION 5

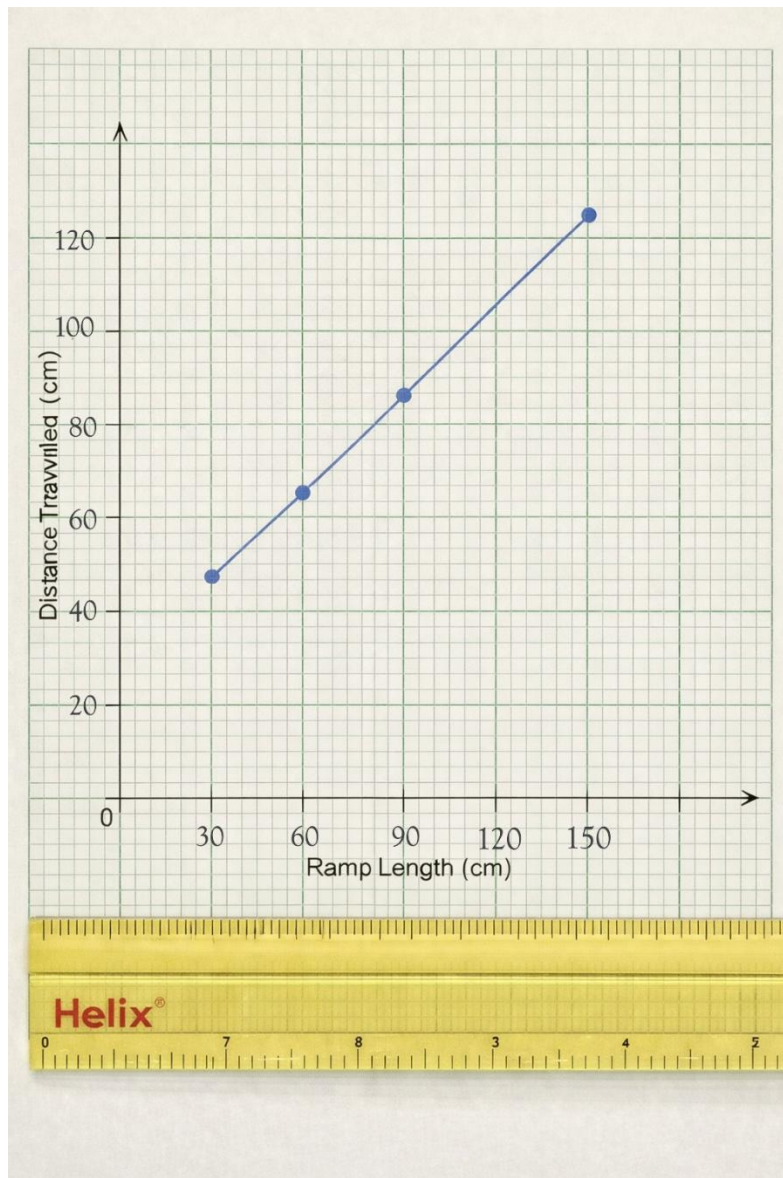
DRAWING A GRAPH

Graph Checklist

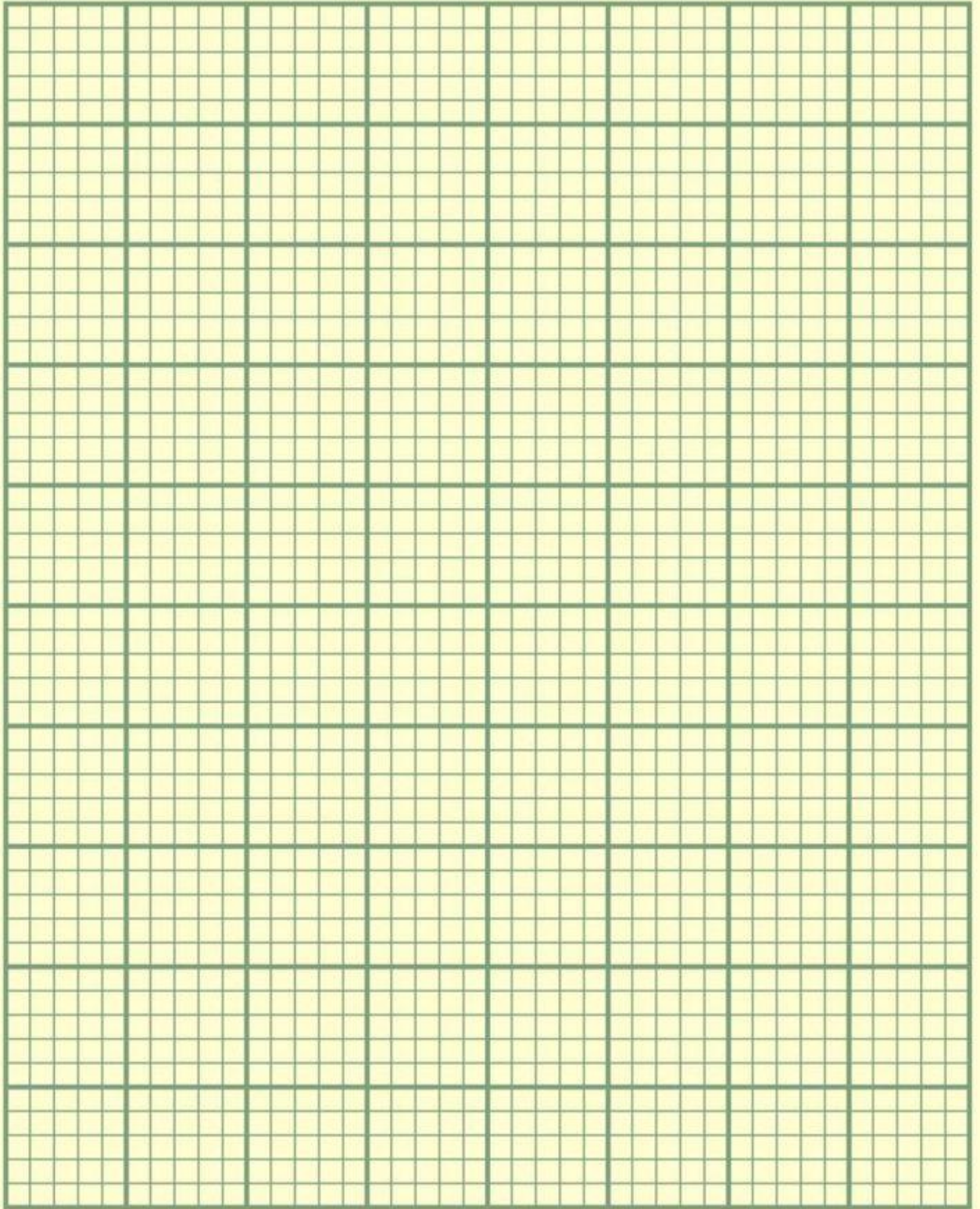
Before you draw your graph, decide and check the following

- ☐ Title
- ☐ Labelled axes
- ☐ Correct units
- ☐ Sensible scale

 example graph (sample)



5 Squares/Inch Engineering Graph Paper



SECTION 6

CONCLUSION

A conclusion answers the question using evidence.

Sentence Starters (use one or more):

- The results show that...
- As the independent variable increased...
- This means that...

 **My Conclusion:**

SECTION 7

EVALUATION & IMPROVEMENT

Scientists always think about how their investigation could be better.

What Worked Well?



What Could Be Improved?



One Improvement I Would Make Next Time



FINAL REFLECTION: BECOMING A SCIENCE INVESTIGATOR

Complete the sentences below:

- I learned that science is about _____

- I can now explain what a fair test is because _____

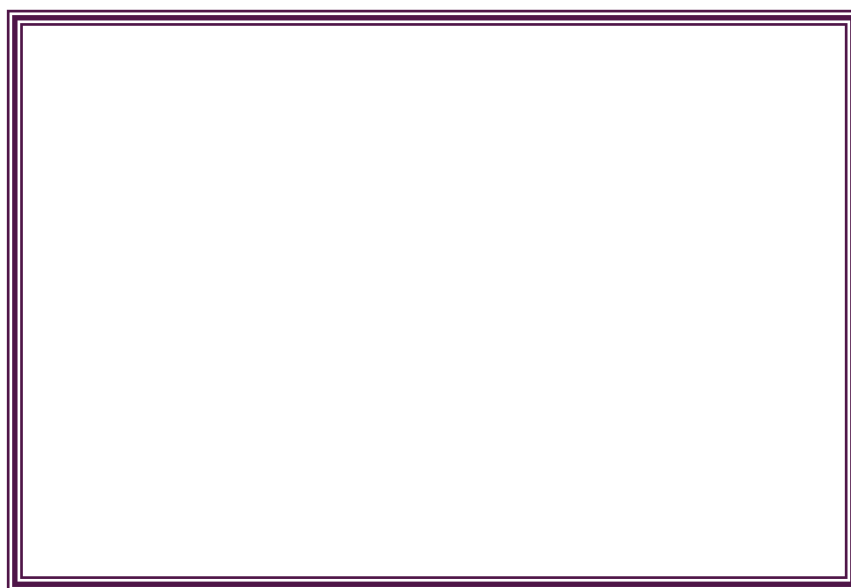
- One thing I am proud of in this project is _____

PROJECT 1 BADGE

You earn this badge when you can:

- plan a fair test
- identify variables correctly
- record data carefully
- explain results using evidence

This Science Investigator Badge is awarded to:



The great Science Investigator

Awarded for thinking carefully, working fairly, and using evidence.

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Well done.

Congratulations! You have completed **Project 1, The Great Science Investigator.**

You are no longer just doing experiments. You are thinking like a scientist!

See you on our Next projects!

Great Job! Get yourselves a nice cupcake 😊

Project Sheet

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Project 1: Ramp Length Investigation

Investigation Question

How does the length of a ramp affect how far a toy car travels?

What You Will Need

- A **toy car** (or small object that rolls)
 - **Books** (to make a ramp)
 - A **flat surface** (floor or table)
 - A **ramp** – anything you can use as a ramp
 - A **ruler or tape measure**
 - **Graph paper**
 - Pencil
-

Very Important Rule (Fair Test)

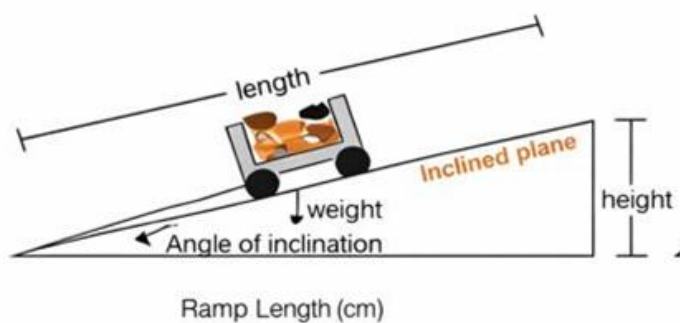
The **height of the ramp** must stay the same for every test.
Only the **length of the ramp** is allowed to change.

⚠ Do **not** push the car — let it roll by itself.

Method (What to Do)

1. Stack some **books** to make a ramp.
2. Place the **top of the ramp** on the books.
3. Use these **suggested ramp lengths**:
 - **30 cm**
 - **60 cm**
 - **90 cm**
4. Each time, move the **bottom of the ramp** further away.
5. Make sure the **books stay the same height** every time.
6. Place the car at the top and let go gently.
7. Measure how far the car travels - Measure the distance from the bottom of the ramp to where the car stops.
8. Record your results in the table.

(If you cannot use these exact lengths, choose three different lengths and write them clearly.)





Note:

- for this experiment, the **books stay the same height**
- Only the **ramp length changes**

Results Table

All distances are measured in Centimeters (cm).

	Trial 1	Trial 2	
Ramp Length (cm)	Distance Travelled (cm)	Distance Travelled (cm)	Average Distance Travelled (cm)
30			
60			
90			

Graph

- Use **graph paper**
- **X-axis:** Ramp Length (cm)
- **Y-axis:** Distance Travelled (cm)
- Use a ruler to draw straight axes.
- Plot your results neatly

Remember:

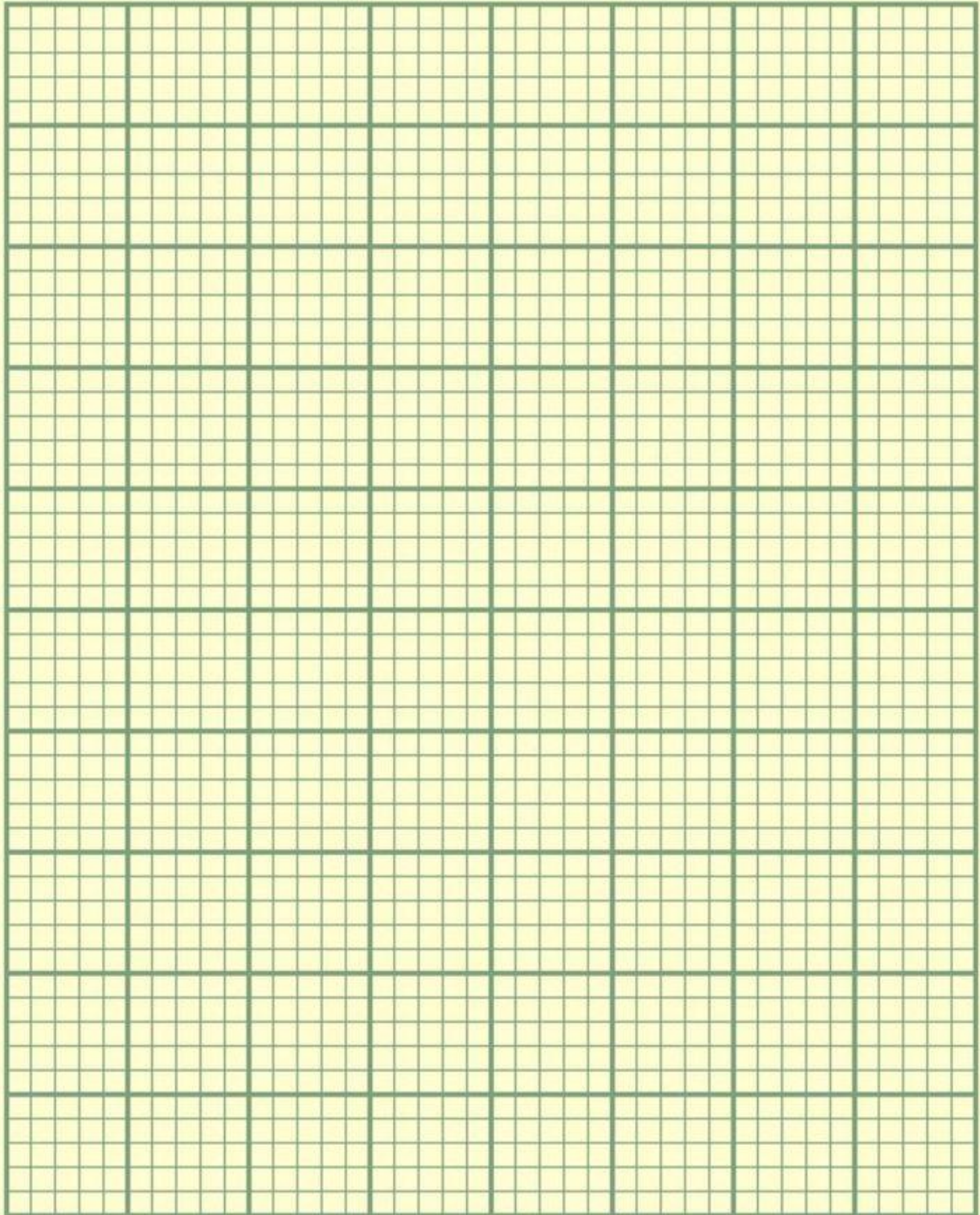
- to label axes clearly
- to choose sensible scales (e.g. 1 big square = 10 cm)

Science Check ✓

- ☐ Same ramp height every time
- ☐ Same car used
- ☐ No pushing
- ☐ Table completed
- ☐ Graph labelled correctly

Graph

5 Squares/Inch Engineering Graph Paper



Conclusion

Write a short paragraph answering:

- What happened as the ramp got longer?
- What pattern did you notice?
- What does this tell you?

Sentence starter (optional):

As the ramp length increased, the distance the car travelled...

This image shows a full page of white paper with horizontal blue dashed lines. The lines are evenly spaced and run across the entire width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

Challenge

Why do you think changing the ramp length affects how far the car travels?

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Learning how science works

By
E. K. Appiah

Science is not about getting the “right answer.”

It is about asking good questions,
testing ideas fairly,
measuring carefully,
and explaining what the evidence shows.

In this workbook, you have taken your first steps
into thinking like a scientist.

In Project 1, pupils learned to:

- Ask scientific questions
- Plan and carry out a fair test
- Identify variables
- Record and analyse data,
- Draw and interpret graphs
- Explain conclusions using evidence



Awarded for thinking carefully,
working fairly, and using evidence.

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